

RAJALAKSHMI ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna University

Rajalakshmi Nagar, Thandalam – 602 105

Web Technology And Mobile Application

Experiment 8

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By

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AIM:

To develop a **web-based Online Railway Reservation System** that validates user input using JavaScript and performs **server-side processing using Servlet**, storing reservation details in a database using JDBC.

ALGORITHM:

1. Create a **Dynamic Web Project** in Eclipse.
2. Design a **JSP page (exp8.jsp)** with a reservation form.
3. Implement **JavaScript validation** to check required fields and valid inputs.
4. Create a **Servlet (RailwayReservationServlet.java)** to handle form submission.
5. Perform **server-side validation** in the servlet.
6. Establish **database connectivity using JDBC**.
7. Create a **MySQL database and reservations table**.
8. Insert user data into the database using PreparedStatement.
9. Display a **styled success/failure response page**.
10. Deploy the project on Apache Tomcat server and run using localhost.

CODE:

Servlet code:

```
package wtma;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.io.IOException;
import java.io.PrintWriter;

@WebServlet("/railwayPreferences")
public class exp7 extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
```

```

response.setContentType("text/html");
PrintWriter out = response.getWriter();

String theme = "Light";
String language = "English";

// Retrieve cookies
Cookie[] cookies = request.getCookies();
if (cookies != null) {
    for (Cookie c : cookies) {
        if (c.getName().equals("theme")) {
            theme = c.getValue();
        }
        if (c.getName().equals("language")) {
            language = c.getValue();
        }
    }
}

// HTML + CSS Output
out.println("<html><head><title>Railway Reservation
Preferences</title>");
out.println("<style>");
out.println("body { font-family: Arial; text-align: center; background-
color: "
    + (theme.equals("Dark") ? "#222; color:white;" : "#f2f2f2;
color:black;") + "}");
out.println(".box { margin-top: 50px; padding: 20px; background:
white; border-radius: 10px; display:inline-block;}");
out.println("</style>");
out.println("</head><body>");

out.println("<h1>  Online Railway Reservation System</h1>");

out.println("<div class='box'>");
out.println("<h2>Your Preferences</h2>");
out.println("<p><b>Theme:</b> " + theme + "</p>");
out.println("<p><b>Language:</b> " + language + "</p>");

out.println("<h3>Update Preferences</h3>");
out.println("<form method='post'>");

```

```
out.println("Select Theme: ");
out.println("<select name='theme'>");
out.println("<option value='Light'>Light</option>");
out.println("<option value='Dark'>Dark</option>");
out.println("</select><br><br>");
```

```
out.println("Select Language: ");
out.println("<select name='language'>");
out.println("<option value='English'>English</option>");
out.println("<option value='Hindi'>Hindi</option>");
out.println("<option value='Tamil'>Tamil</option>");
out.println("</select><br><br>");
```

```
out.println("<input type='submit' value='Save Preferences'>");
out.println("</form>");
out.println("</div>");
```

```
out.println("</body></html>");
```

```
}
```

```
protected void doPost(HttpServletRequest request,
HttpServletRequest response)
    throws ServletException, IOException {
```

```
String theme = request.getParameter("theme");
String language = request.getParameter("language");
```

```
// Create cookies
```

```
Cookie themeCookie = new Cookie("theme", theme);
Cookie languageCookie = new Cookie("language", language);
```

```
// Set expiry (1 day)
```

```
themeCookie.setMaxAge(24 * 60 * 60);
languageCookie.setMaxAge(24 * 60 * 60);
```

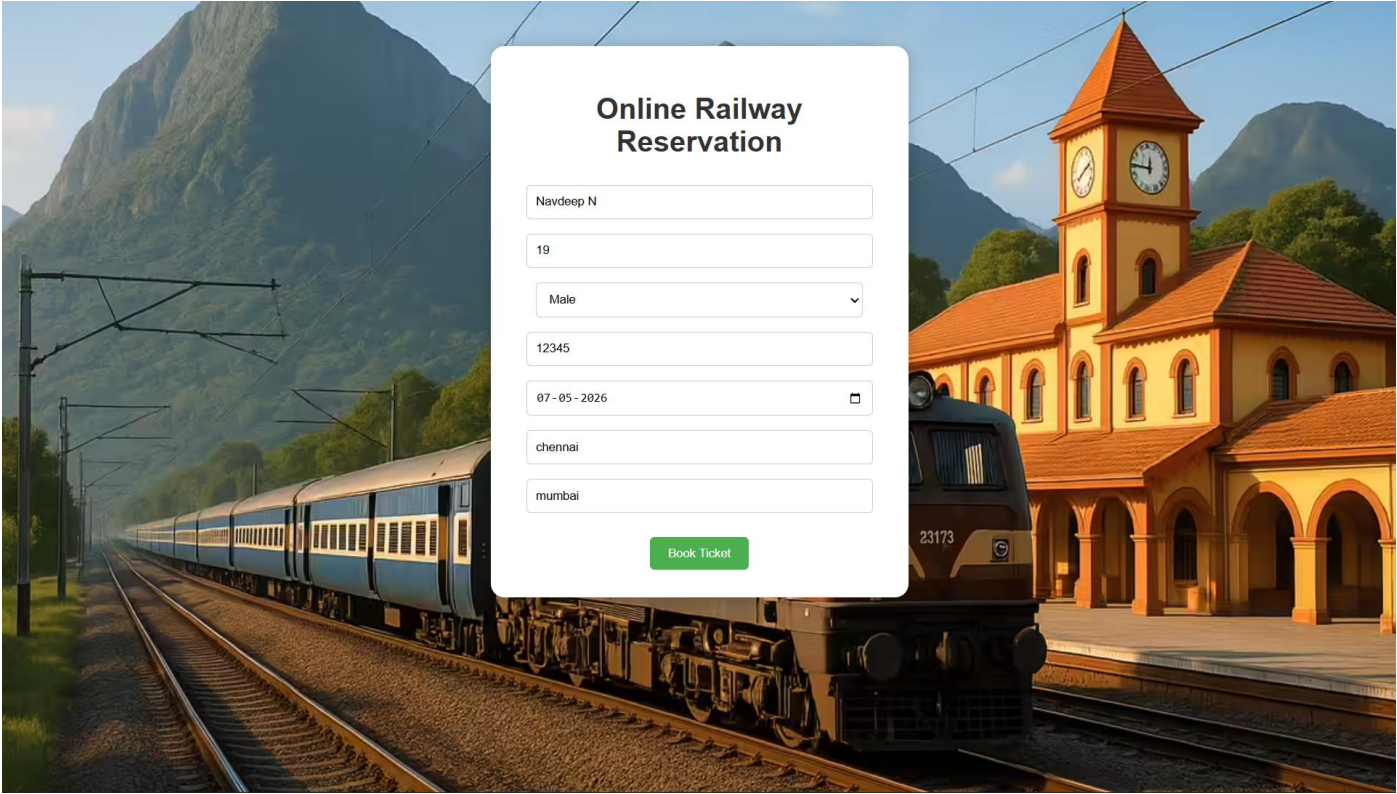
```
// Add cookies to response
```

```
response.addCookie(themeCookie);
response.addCookie(languageCookie);
```

```
// Redirect back to GET to display updated values
```

```
    response.sendRedirect("railwayPreferences");  
  }  
}
```

OUTPUT:



Online Railway Reservation System

Admin Dashboard

System Status and Updates

✓ Success! Booking reference table has been updated.
Update timestamp: 2023-11-15 14:30:00 PST

Current Booking Period: LIB-1234 (Active)
Total Records Updated: 100,456
System Status: OPTIMAL
Last Full Backup: 2023-11-14 23:59:00 PST
Refresh System Status

Recent System Activities

Time	Activity	Status
2023-11-15 14:30:00	Table Update:	Success
2023-11-15 13:45:12	Cache Clear:	Partial
2023-11-15 12:00:05	User Audit:	Success

© 2023 Rail Reservation Solutions Inc. | System Status: Optimal

RESULT:

The Online Railway Reservation System was successfully implemented. User inputs were validated on both client and server sides, and reservation details were **stored in the MySQL database**. A **styled output page** displayed the reservation status correctly.